



Quality Assurance Guidelines Prototype IV Lidl US



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1. Abbreviation



Acceptable



Not Acceptable



References to the drawings or specification

HQ

Headquarter

CM

Construction Manager

QM

Quality Manager

QA

Quality Assurance

GC

General Contractor

RTU

Roof Top Unit

CU

Cooling Unit

ATS

Automatic Transfer Switch

2. Intro

2.1 General Information

Main purpose of the quality assurance guideline is to meet all the requirements from a qualitative and contractual point of view that will help to ensure the construction process will be complete on time and without defects.

2.2 Purpose

Quality Assurance Guidelines are designed with the goal to achieve a “Zero Punch List” program that will lead to the lower operation cost in the future. Guidelines should minimize number of mistakes done during the construction of Lidl Stores and ensure that the completion of the project will be on time and on budget.

This guideline covers topics where most of the mistakes and issues are taking place during the construction, hence it should help to prevent them in the future and build future stores – more efficient and sustainable.

The information used in this guideline was obtained during the construction of the test store, located in Aberdeen in Maryland. Due various reasons such as climate or local jurisdiction - some of construction solutions or applications will vary from case to case

3. Site Development and Earthwork

This chapter shall be applied as a general guide to all type of soils and foundation work. All the work shall be performed in accordance of applicable building codes and current technical and structural documentation related to the specific site

If unusable soils exist and/or non-prototype foundations are required, please utilize the local design, engineering and consultants to provide appropriate design and methods

3.1 Survey and property lines

Prior to construction – site specific approved plan must be present at the jobsite. This include at a minimum, current plans: Survey plan, Geotech/third party inspection scheduled and passed, passed soil pressure capacity inspection, full building and site plans

NOTE: Reference: S101 - Foundation Schedule, Note 2

Property lines and construction limits shall be subdivided according to approved site plan. Silt fence and property fence shall be adjusted during the construction and remain until the end of the project or the stage.



3.2 Before you dig – Check Utility or call 811

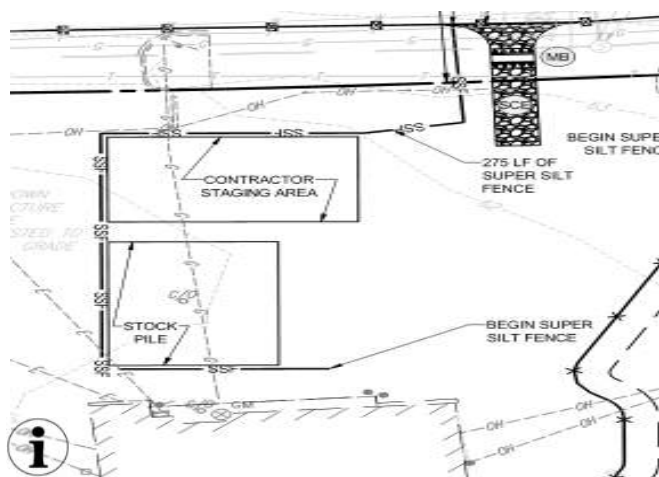


Prior to any type of earthwork all the utilities must be identified and marked. All utilities that require cleanout need to be located and marked with a permanent sign.



Any exposed/unknown utilities must be capped off or properly abandoned, after verification that the utilities are no longer in service

3.3 Site logistics - staging and working area





Note

1. Staging and work areas (parking, restroom areas, etc.) must comply with local jurisdiction and installed according to approved layout to maximize the efficiency of construction traffic.
2. In accordance with the local jurisdiction – all the approval notices/Permits must be posted and remain on the site for the duration of the project

4. Grading and excavation

4.1 Grading

Main purpose of grading is to alter the land surface to make it compatible with the intended use, establishes and controls the new drainage patterns, helps define the character and aesthetics of the site.



4.2 Prior to breaking the ground and during the excavation

To ensure correct execution of the trenching – survey team must be present on the site during the excavation and after the concrete pour



Note

All the paint markings shall remain on the ground and must be verified before excavation. Location and the direction of the footings must be with accordance to the site plan(location) and building plans (design)

5. Structure

5.1 Rebar installation

After the excavation and prior to installing the reinforcements trenches must be free from any debris and standing water



All the rebars must be elevated from the ground and comply with the building norms. All the spacers shall be installed prior to the concrete is poured

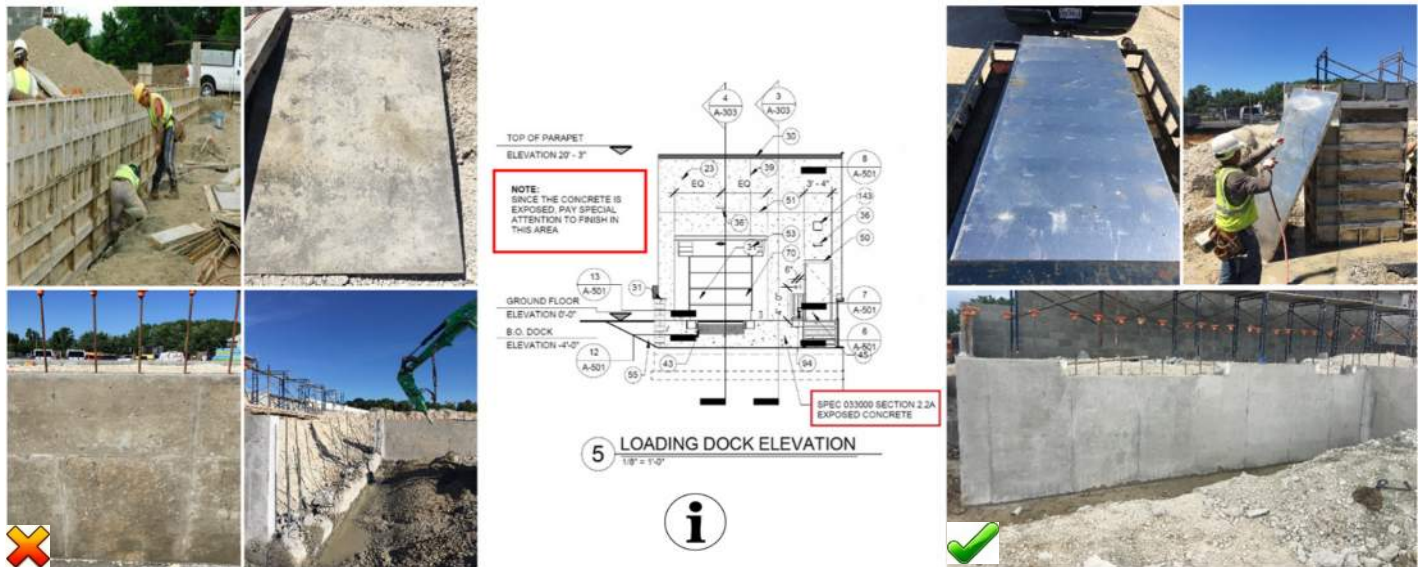
Example. 1 Rebar installation with future anchoring



Column baseplate installation per structural drawings page S-201 (Detail 25,26)

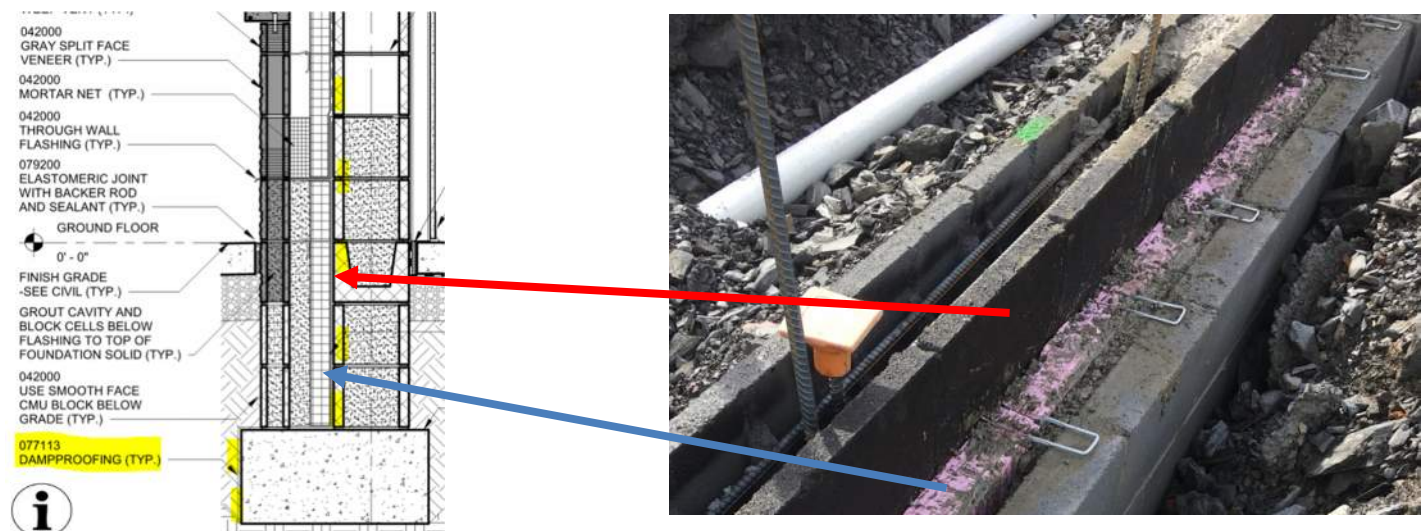
5.2 Concrete walls formwork

Concrete wall forms walls must be cleaned and prepared to apply release agent



5.3 Damp proofing, vapor barrier and insulation

The goal of damp proofing is to prevent moisture from passing through the walls into the interior space



Damp proofing must be installed per approved drawings and shall be installed to the acceptable level. No empty spots shall be open and not covered.



Damp proofing should be applied in a way, so the top of the proofing will be covered or overlapping with the through wall flashing

5.4 Vapor Barrier

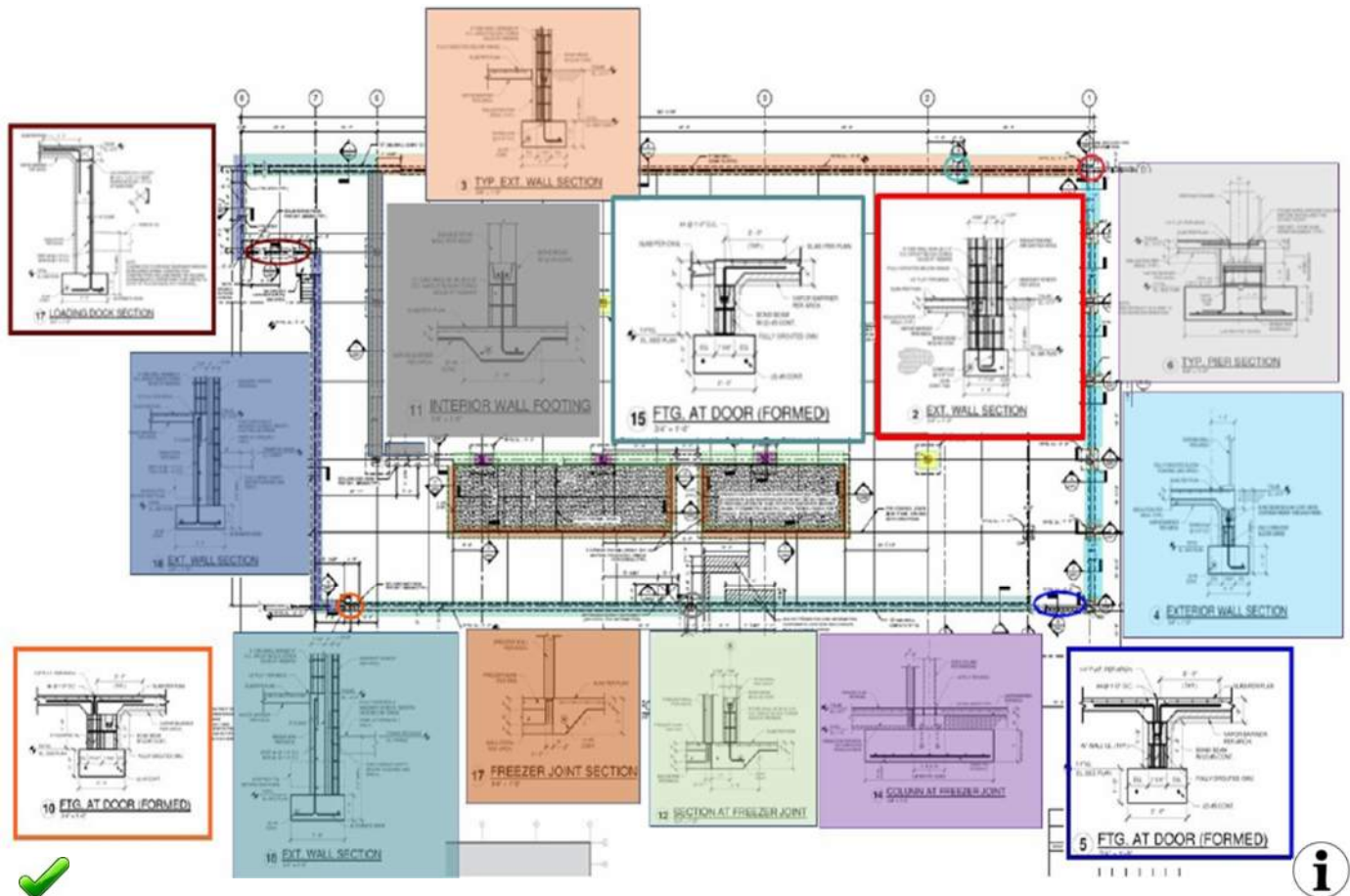
The function of a vapor barrier is to protect the concrete mixture from water lost that could be absorbed by soils. Also, it will help to retard the migration of vapor. Where it is located in an assembly and its permeability is a function of climate, the characteristics of the materials that comprise the assembly and the interior conditions.



5.5 Insulation

Main purpose of the insulation is to provide consistent temperature control during a cold and hot period. Additionally, a well-insulated building assists in controlling the operational cost of the store. Therefore, different areas of the store are designed with different type and different levels of the insulation.

The various plans below detail multiple applications utilized on the prototype 4 store. Different variation of the insulation will be advised for each climate area by local jurisdictions

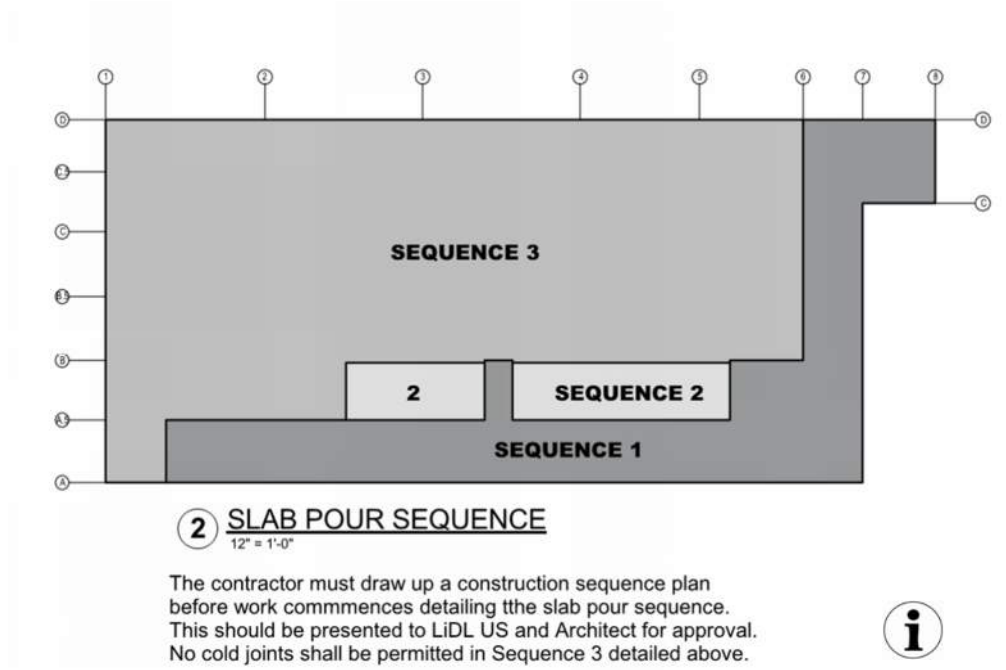


Warning

Any deviation from the approved drawings are not permitted. All adjustments to the structure must be done at GC cost

5.6 Slab

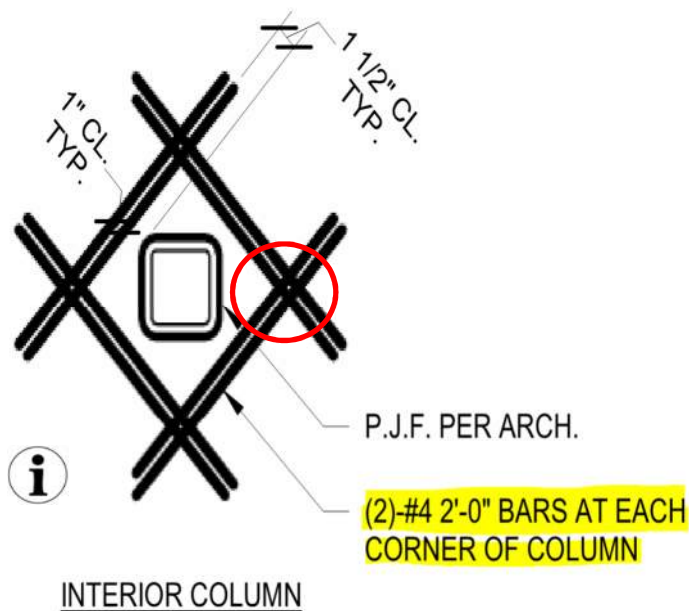
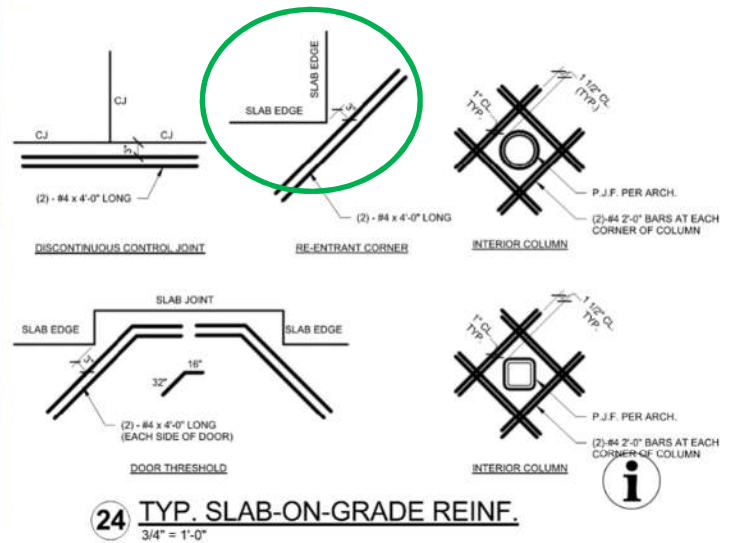
The “Slab” has to be installed in the sequence that was established on the approved drawings. All under slab MEP systems or conduits shall be installed and secured prior to concrete has been poured (see Part 5 “MEP Systems”)



Each sequence area must be enclosed prior to the pour



Slab reinforcement shall be installed per drawings and checked by CM with the checklist (see checklists Foundation, Slab)



Warning

Reinforcement and embedded items are NOT configured in accordance with applicable approved shop drawings or plan details.

5.6.1 Concrete pour

Concrete shall be constructed in accordance with the building design and engineering drawings. Quality control must be provided during and after each pour. Field test must be performed before each pour.

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FIELD REPORT

Work Order No.: WOHAG18002605	Date: December 12, 2018	
HCEA Project No.: H18108	Report No.: 65	Weather/Temp: Partly Sunny 39 °F
Project Name: LIDL-Hagerstown	On Site Time: 11 hr	Travel Time: 0.5 hr
Client: Lidl US, LLC	Total Time: 11.5 hr	
Contractor: Doyle Construction		

Description of Work:
 HCEA representative arrived on site to provide testing and observation services.
 Landmark Concrete LLC poured the 4" thick back of house floor slab in the morning.
 Wire chairs were in place supporting the welded wire mesh before pour started.
 Observed the placement of approximately 121 cubic yards of 4000 psi concrete by hand.
 The concrete tested was in general conformance with the project specifications at placement time.



GC shall follow a Good Practice during the concrete pour – tranches must be free of litter and present no earth cavities, no objects that were not designed to stay in the area shall be removed



Picture shall be used as a reference – footers area is shown



Warning

Example: Prior the footing installation all the earth cavities must be covered

5.6.2 Concrete slab (Section 033010-3.10)

After the placement the concrete, GC shall protect the concrete from premature drying, excessively hot or cold temperatures, and mechanical damage.



GC shall maintain concrete slab protection for the duration of the project to prevent damage to the floor surface.



5.6.3 Slab Joins (Section 033010-3.8)

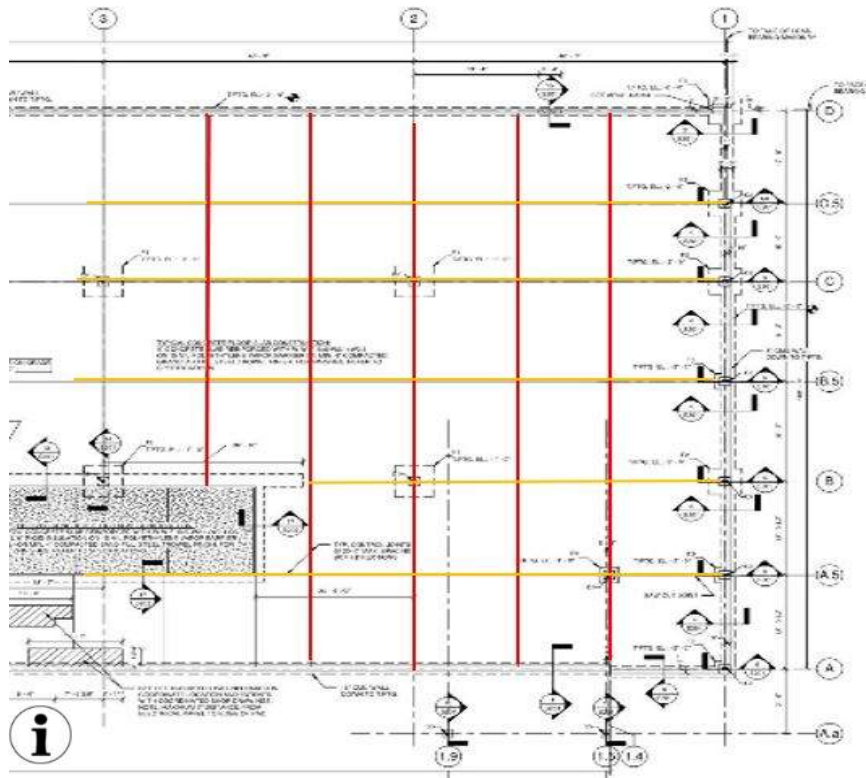
Main purpose of the joints is to control appearance of the cracks on the surface of the floor, and to control their direction due to physics of the concrete during the curing process.



GC Shall use a new cutting saw with equipment that is set in the appropri-



The cuts should be made at a predetermined spacing and only after the concrete has obtained sufficient strength but before internal cracking begins. Therefore, the timing of saw cuts is critical. Determining when to make cuts depends on many factors, including the type of admixture and aggregate used in the concrete, the air temperature, and the concrete's mix design.



Example of the control joints on the concrete slab

5.6.4 Concrete finish (Section 033010-3.7)

Concrete floor shall be installed with a dense, uniform in color and profile finish. Since, different areas in the store requires different type of application. (see info below)

ROOM NO.	ROOM NAME	Area	FLOOR		BASE	
			MAT'L	FIN	MAT'L	FIN
100	VESTIBULE	325 SF	CONC	F1	B1	FF
101	SALES	15456 SF	CONC	F1	B1	FF
102	BAKERY	610 SF	CONC	F5	B2	SS
103	UNISEX RESTROOM	89 SF	CONC	PT	B1	FF
104	UNISEX RESTROOM	89 SF	CONC	PT	B1	FF
105	AMENITY HALLWAY	115 SF	CONC	F1	B1	FF
106	JANITOR CLOSET	15 SF	CONC	F3	B1	FF
107	OFFICE	119 SF	CONC	F3	B1	FF
108	OFFICE	117 SF	CONC	F3	B1	FF
109	BREAK ROOM	453 SF	CONC	F3	B1	FF
110	IT ROOM	112 SF	CONC	F2	B1	FF
111	TECHNICAL ROOM	339 SF	CONC	F3	-	-
112	BACKSTOCK	2018 SF	CONC	F3	-	-
113	SPRINKLER RISER	59 SF	CONC	F3	-	-
114	STAGING	1972 SF	CONC	F3	B1	FF
115	LOADING DOCK	426 SF	CONC	F3	B1	FF
116	COOLER	498 SF	CONC	F5	-	-
117	FREEZER	547 SF	CONC	F6	-	-
118	BAKERY FREEZER	890 SF	CONC	F6	-	-
119	SCRUBBER	36 SF	CONC	F4	-	-

FLOOR FINISHES		
FLOOR FINISHES	F1	POLISHED CONCRETE
FLOOR FINISHES	F2	PAINTED CONCRETE
FLOOR FINISHES	F3	SEALED CONCRETE
FLOOR FINISHES	F4	METAL GRATE
FLOOR FINISHES	F5	CLEAR SLIP RESISTANT COATING (EPOXY)
FLOOR FINISHES	F6	SLIP RESISTANT COATING (EPOXY)



Therefore, GC should follow the drawings and specification to apply appropriate finishes with required amount.



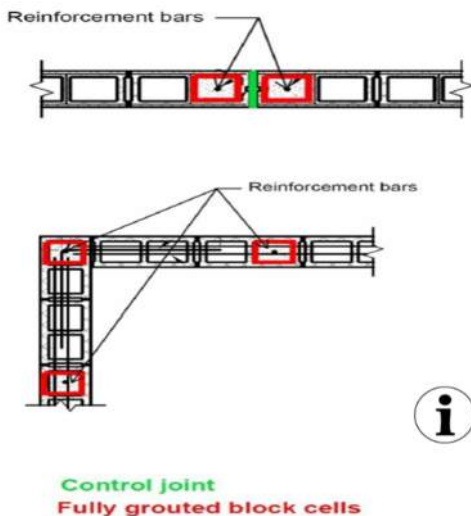
5.7 Masonry (Section 042223, 042000)



For the duration of the construction masonry units shall be stored on the pallets to prevent damage and extensive moisture. GC should not use any damaged or that deviates from the approved color or shape.

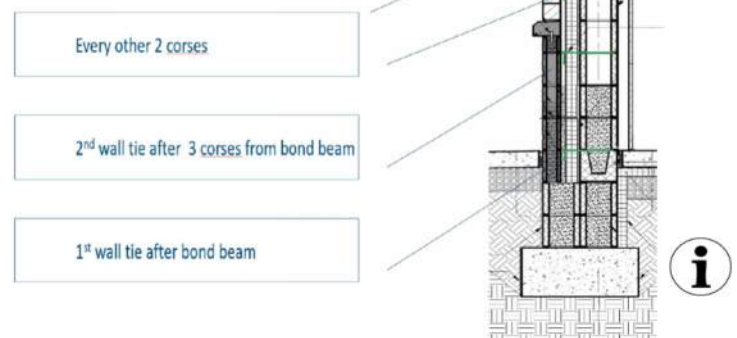
5.7.1 Wall ties and masonry reinforcement (Section 042000-3.7,3.10)

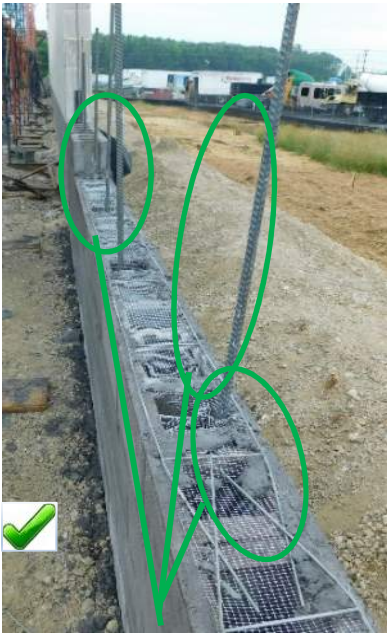
Wall ties perform three primary functions between a **wythe** of brick and its backing or another wythe of masonry: 1) provide a connection, 2) transfer lateral loads, 3) permit in-plane movement to accommodate differential movements and, in some cases, restrain differential movement. In addition to these primary functions, metal ties (as joint reinforcement) may also be required to serve as horizontal structural reinforcement or provide longitudinal continuity.



• Wall Ties

(used to tie the internal and external walls constructed of bricks, making the two parts to act as a homogeneous unit)

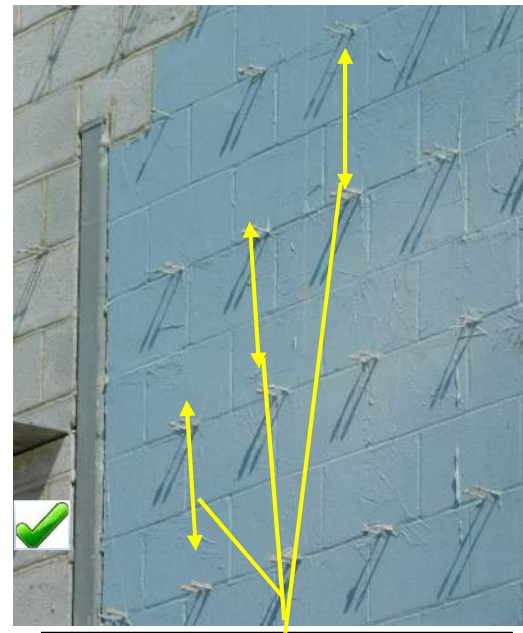




Joint and vertical reinforcement is installed per specification



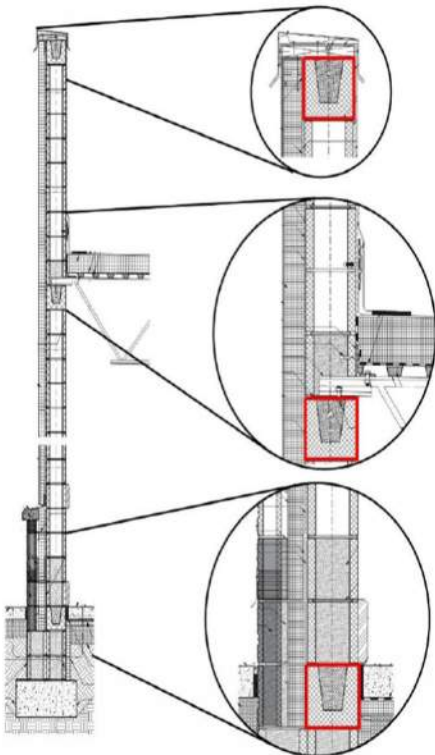
Reinforcement bars for the control/expansion joints installed. Block cells are fully grouted



Unit Ties - Installed at equal distance

5.7.2 Bond beam

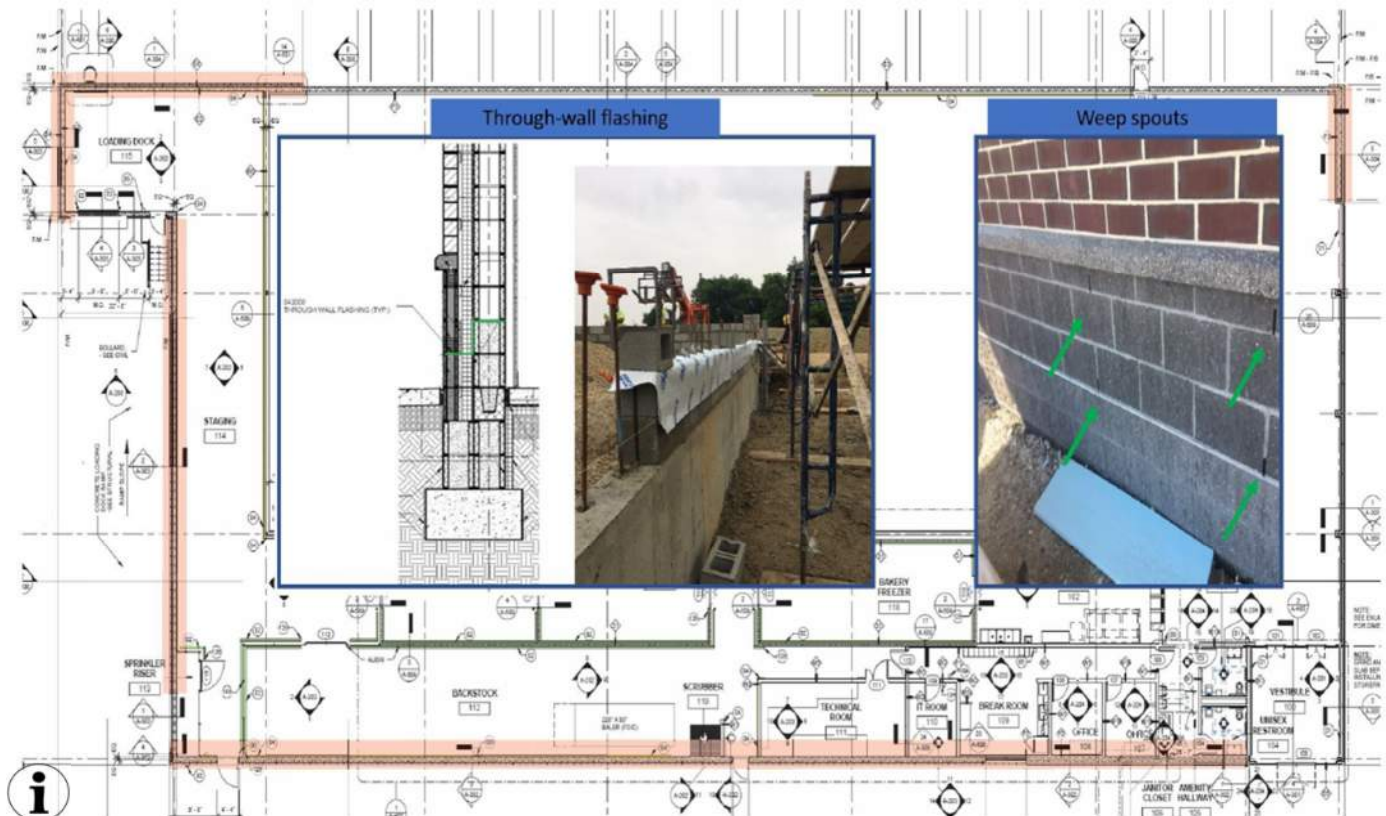
Bond beam shall be installed as noted on the drawings and specification. Units are permitted to be used as it designated, no modification to the standard CMU units shall be permitted



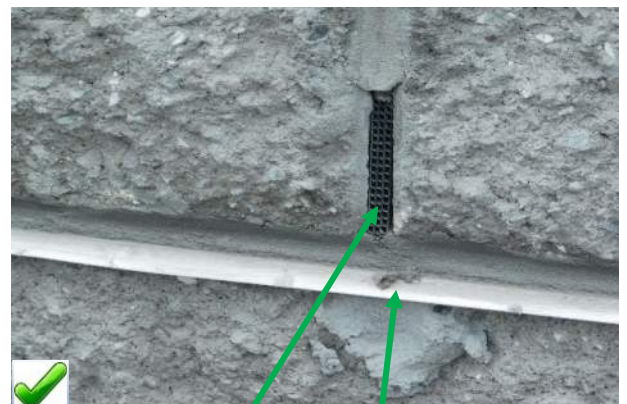
5.7.3 Moisture control in cavity type walls

Even with good building practices, some water penetration will occur. This is where flashing helps. The function of flashing is to collect and divert any moisture penetrating the wall or sill.

GC should follow local jurisdiction and prototype specification in order to properly control moisture and water penetrating the structure.



Example of through wall flashing



Weep hole and metal drip edge



Missing through wall flashing



Step down through wall flashing, installed on loading dock area

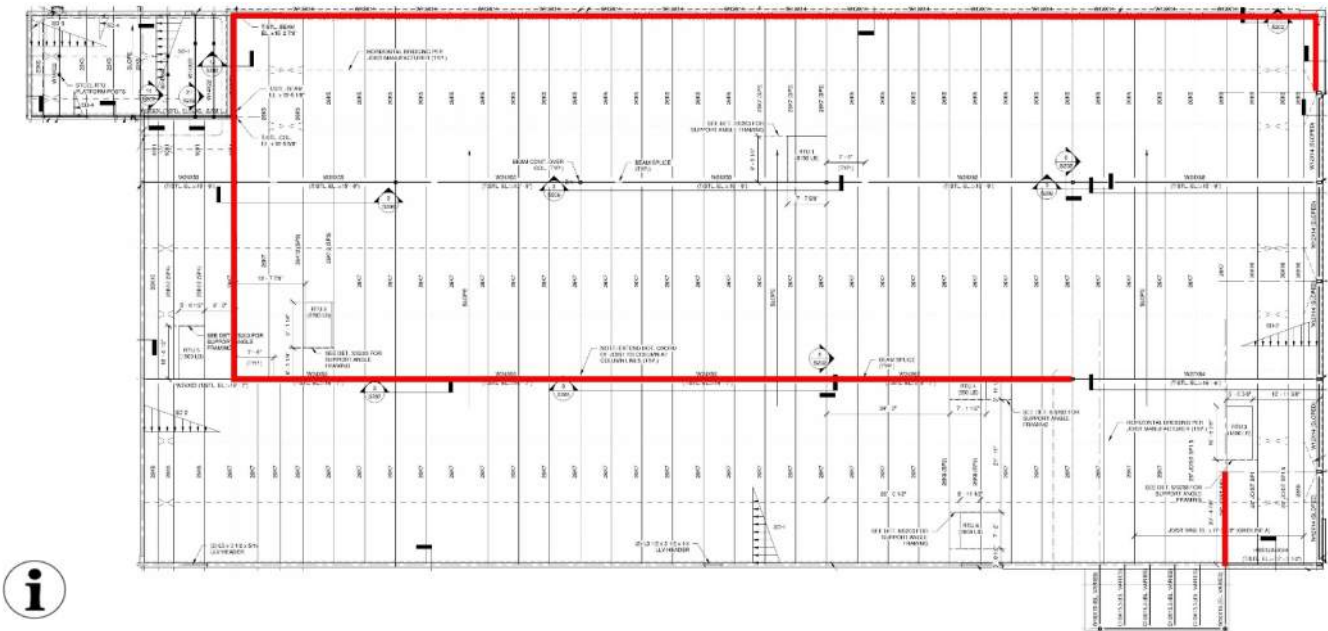
5.8 Steel Structure – Column, Beams, Trusses, Roofing (Sections 051200 - 055313)

Structural steel isn't just for skyscrapers, large agriculture buildings and garages. There are multiple reasons why steel makes an attractive building option: Sustainability, Affordability, Durability

5.8.1 Connections

All connections have to be done with accordance to the approved drawings and specification Section 051200-3.3, and comply with local jurisdiction requirements

In "Prototype 4" stores all concealed connections shall be welded, and all exposed connection shall be bolted





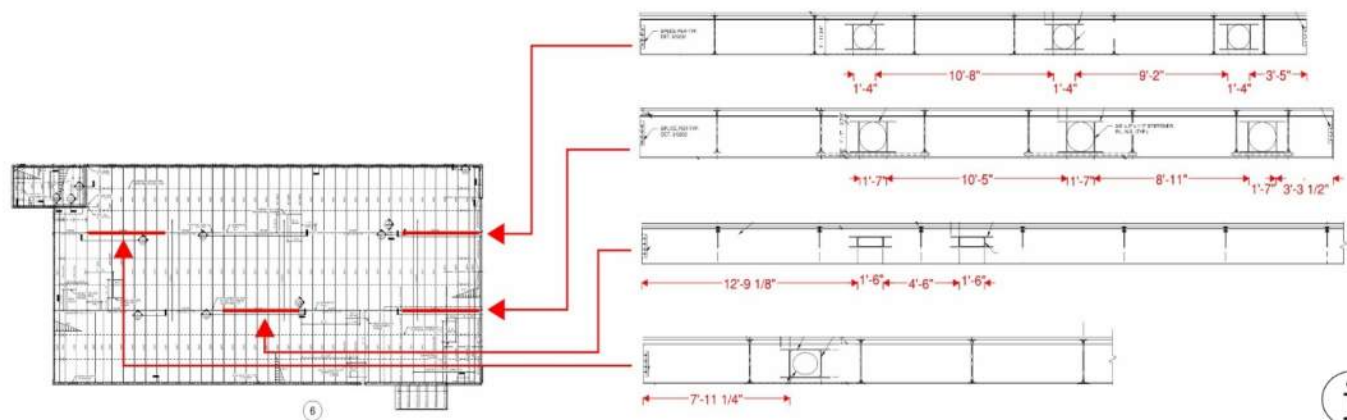
Any deviation from the design should be verified with a third-party inspection



5.8.2 Beams



Beams with penetration for ductwork and electrical conduits shall be installed per drawings. No modification to the opening in the beams shall be done on the field.



All the dimensions for the openings in the beams shall be verified by GC prior to installation, and confirmed by CM



5.8.3 Trusses

Trusses shall be installed per approved drawings, direction of the ductwork shall be accounted by GC prior placing trusses.



Opening at the location where trusses are supported on the CMU wall shall be grouted

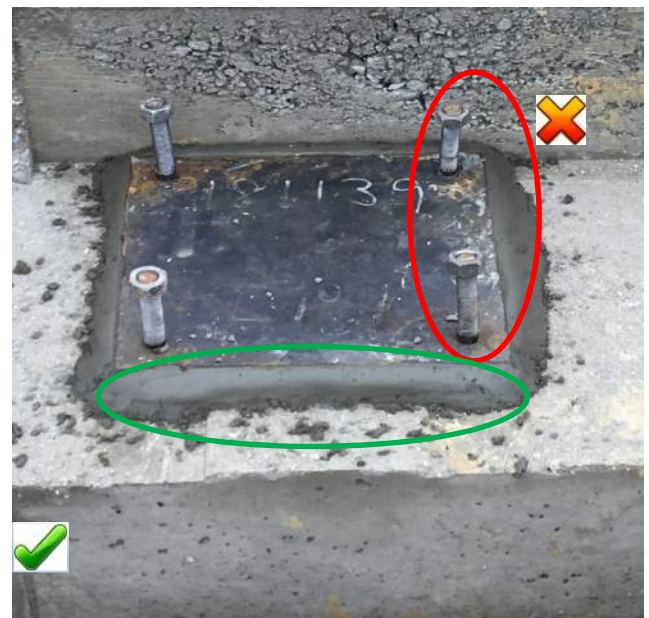


5.8.4 Column

Column must be installed on the base – level and position of which must be verified prior to erection of the column



Each base plate location shall be verified with the surveyor to ensure the correct location and alignment



Each base plate shall be grouted, and bolt connection shall be aligned

After the complete structure is in place all the connection must be tightened, alignment of the columns must be confirmed



All the steel elements arrive at the job site grouted and ready to be placed. GC shall make all the possible precautions to protect the grout layer. All the structures shall be installed free of dirt or concrete. Especially this important for the areas that will not be painted.

Damaged grout layer will increase the corrosion of the structure



6. MEP – Plumbing, Electrical, HVAC

Prior to the beginning of the work – GC and CM shall confirm all approved drawings are present at the job site. All the coordination must be done to avoid conflicts between MEP systems (power lines, ducts, sprinkler systems...)

During the construction each of the system must be verified by CM with the help of Checklist app

6.1 Plumbing

6.1.1 Underground

Prior to concrete pour all underground utilities - plumbing and conduits shall be installed per approved plans



Wrong location of the vent for scrubbing station drain



Correct location of the vent- tie to the pipe under the slab

Prototype 4 stores exists in a different projection (Entrance from the left or from the right) GC must follow the approved drawings

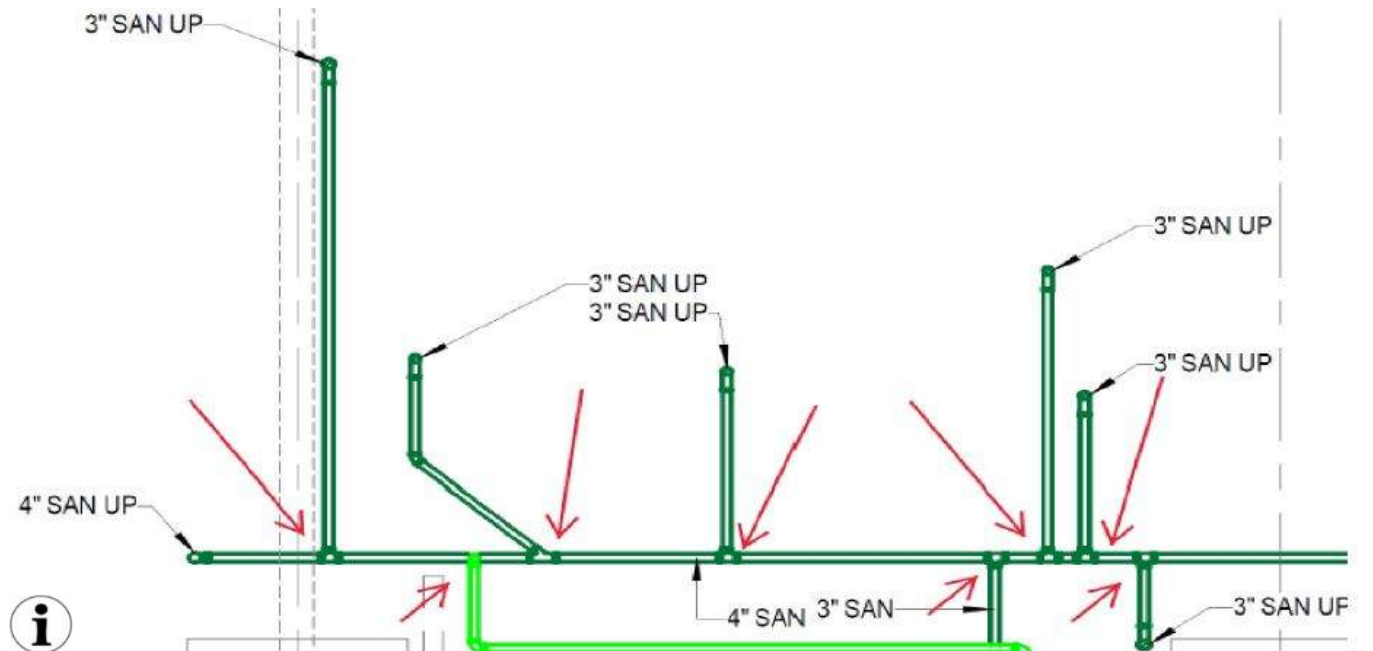


Wrong orientation of the conduits for the freezer - A5 line

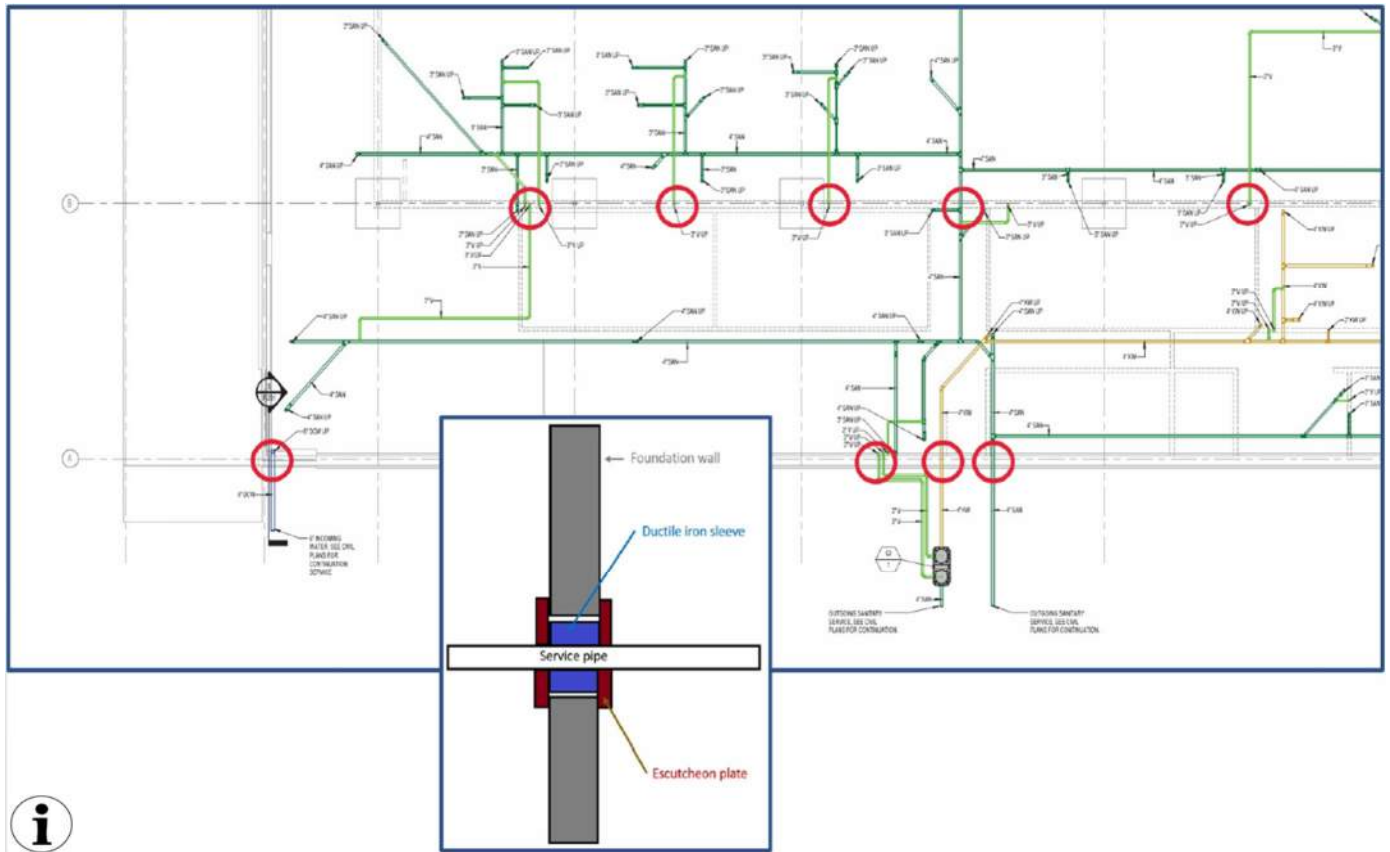


Correctly positioned conduits – B line

CM shall verify if the pressure transition coupling has been installed at joins between two different pipe size (Section 22.13.16.3.3)



For each pipe that designed to penetrate concrete foundation be encapsulated with the iron pipe sleeve (See page P-000, Note 9,10)



GC shall follow specification 26.05.43 section 3.4 – all the pipe clusters shall be spaced with duct bank spacers



Duct/pipe separators shall be placed not less than 5 spacers per 20ft of concrete duct bank – GC shall follow specification 26 05 43 section 3.4 I.1

In order to control expansion and contractions of the ducts – GC shall install expansion fittings for the concrete-encase ducts (see Specification 26.05.43 section 3.4)

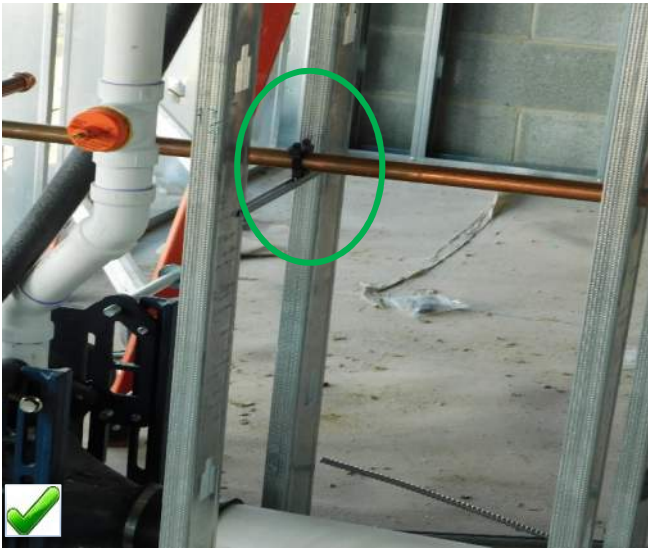
GC shall confirm if all the conduits and ducts are clean and dry – prior to pulling wires. Also, all the duct openings shall remain temporary covered during construction.

All the joints connections hast to be watertight and done with solvent-cemented (See Specification 26 05 43 3.4)



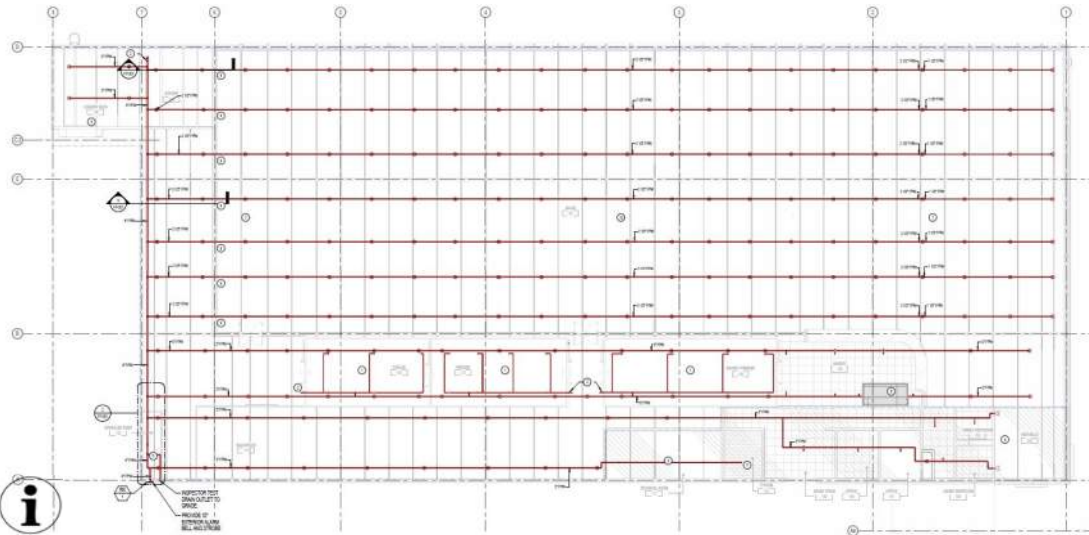
6.1.2 Above ground

All above ground piping should be installed per code or following local jurisdiction



All the piping shall be securely positioned with the accordance to the drawings

All the under-ceiling plumbing – such as sprinkler systems has to be positioned as it shown on the drawings. Deviation are not permitted



Wrong direction of the sprinkler system



6.2 HVAC

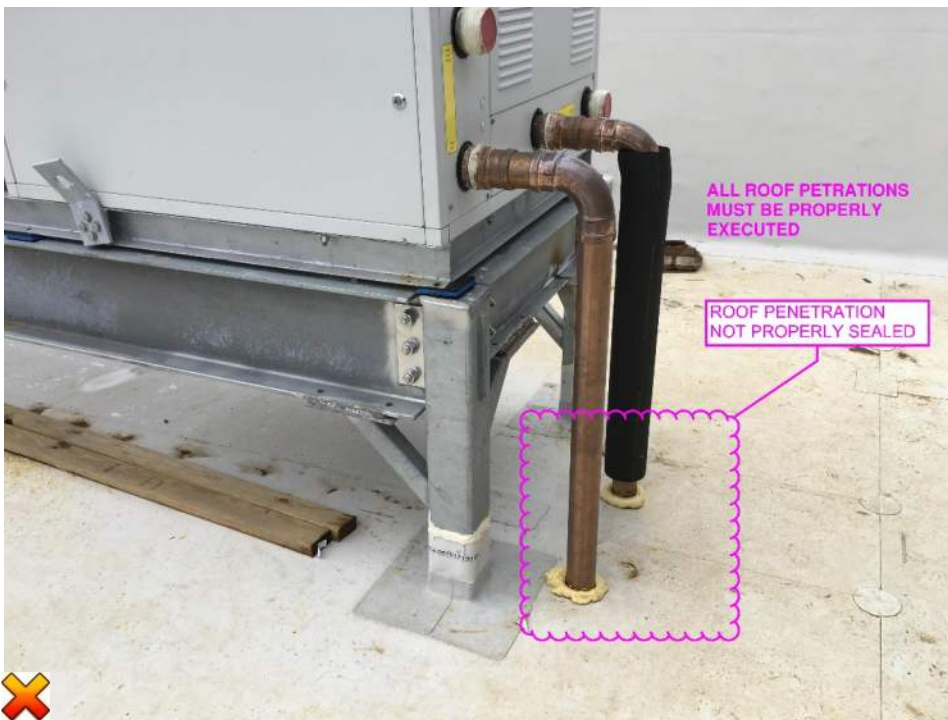
6.2.1 Roof top unit installation

Air handling of the “Prototype 4” stores is done by six (6) rooftop units and two (2) dry cooler. It is crucially important for the quality of the produce and other products to have the units functioning in the proper way, and have the air to be circulated in the way that will be enough for the normal conditions of the store.

Units shall be installed per manufacture recommendation, GC must verify that the unit arrived without any damages.



Any damage to the RTU shall be noted and reported immediately to CM and manufacture.



ALL ROOF PETRATIONS MUST BE PROPERLY EXECUTED

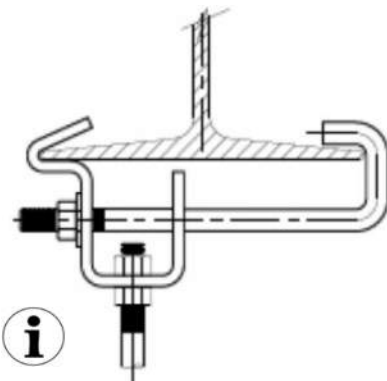
ROOF PENETRATION NOT PROPERLY SEALED

All the roof penetrations must be sealed and encapsulated with steel pipe sleeve (See 23.05.17 section 3.1- 3.3)

All the units must be installed on the curbs, grout, spring/vibration isolators (see specification 23.05.29 section 3.3; 23.34.23 section 3.1)

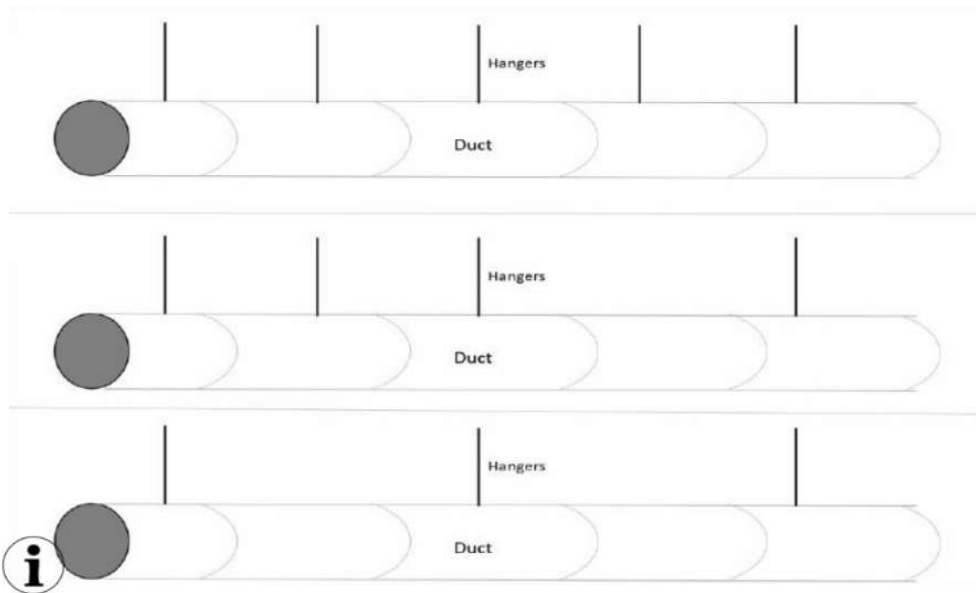


Hinges to support ducts shall be galvanized per specification **23.05.29 section 3.2**, as well as a clamp that are attached to the steel structure



6.2.2 Ductwork

Hinges shall be installed in the way they could distribute the load from the ducts equally and achieve indicated slope (See specification 23.05.29 section 3.5)

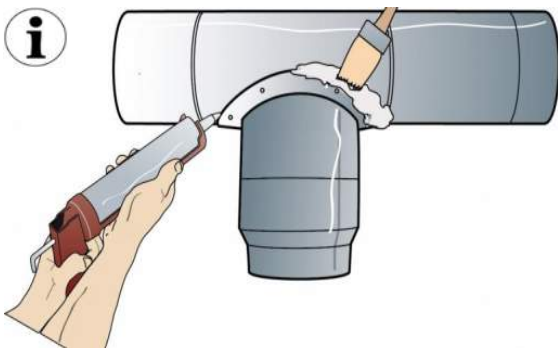


Quantity of hinges will be sufficient - Correct

Hinges are not spaced correctly - NOT correct

Hinges will be under the stress - NOT correct

All the ductwork shall be insulated, and all the connection must be sealed (**Spec 23.31.13**)



6.2.3 Miscellaneous



Rooftop sidewalk and pipe support shall be fully glued to the surface of the roof membrane

6.3 Electrical (Specification 26.05.00 – 26.56.00)

6.3.1 Conduits

Prior to the begging of the construction, GC and CM shall verify if the circuit breaker study is done (See Spec. 26.05.73 section 1.3); if the arc-flash study was complete (See Spec. 26.05.73 section 1.3). Also, GC shall confirm if the shop drawings are on site.

All the conduits with exception of damp and wet areas are EMT (Electrical metallic tubing)

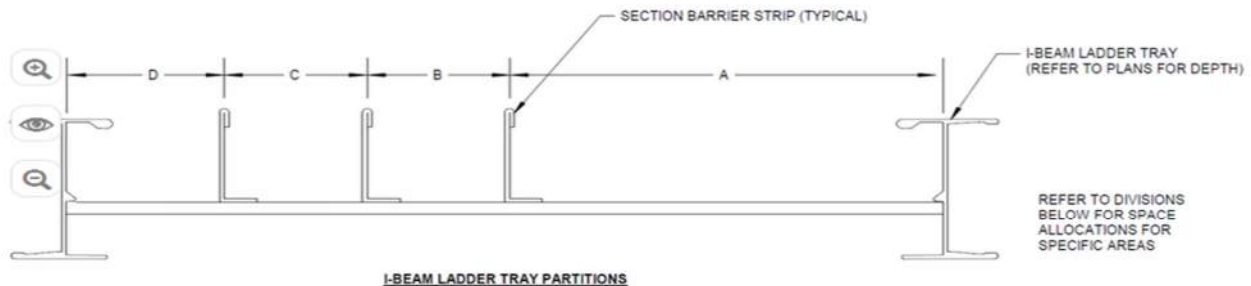


All overhead conduits in the sales area and lobby area shall be installed and tie to the ceiling joists



During the construction all the conduits shall be installed per approved plan, and GC shall verify if all the conduits are parallel or perpendicular to the walls (See E-000 Note 18)

Cable trays shall be installed as it designated per plan, and the cables of each system (power, IT or communication) shall be separated with the barrier strip



DIVISION OF 12" WIDE CLOSED BOTTOM CABLE TRAY (SALES FLOOR AREA):
A - 6" LIDL DATA / SECURITY DATA CABLING / REFRIGERATION CONTROL & BMS / INTRUSION DETECTION SYSTEM CABLING
B - 6" ELECTRICAL CABLING

DIVISION OF 18" WIDE I-BEAM LADDER CABLE TRAY (BACK OF HOUSE AREA):
A - 6" LIDL DATA / SECURITY DATA CABLING
B - 2" REFRIGERATION CONTROL & BMS / INTRUSION DETECTION CABLING
C - 2" QUAIL/MOOD MEDIA CABLING
D - 2" ACCESS CONTROL POWER CABLING
E - 6" ELECTRICAL CABLING

DIVISION OF 12" WIDE I-BEAM LADDER CABLE TRAY (BACK OF HOUSE AREA):
A - 6" LIDL DATA / SECURITY DATA CABLING
B - 2" REFRIGERATION CONTROL & BMS / INTRUSION DETECTION CABLING
C - 2" ACCESS CONTROL POWER CABLING
D - 2" QUAIL/MOOD MEDIA CABLING

DIVISION OF 24" WIDE I-BEAM LADDER CABLE TRAY (BACK OF HOUSE AREA):
A - 12" LIDL DATA / SECURITY DATA CABLING
B - 6" REFRIGERATION CONTROL & BMS / INTRUSION DETECTION CABLING
C - 4" ACCESS CONTROL POWER CABLING
D - 2" QUAIL/MOOD MEDIA CABLING

DETAIL NOTES:

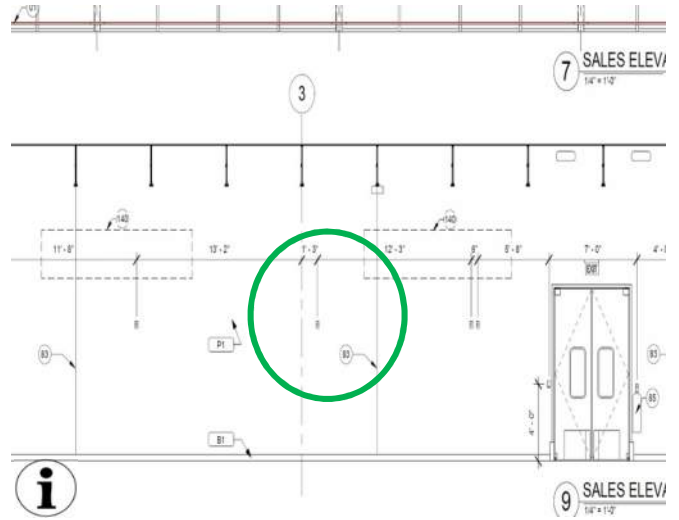
1. REFER TO FLOOR PLANS FOR CABLE BASKET SIZE AND LOCATIONS.
2. CONTRACTOR SHALL FIELD VERIFY I-BEAM LADDER CABLE TRAY SPACERS ALLOWING FOR NO MORE THAN 40% FILL IN EACH SPACE OF CABLE TRAY



CM shall visually verify that there is no damage to insulation, either no sharp corners protuberances in cable trays that will case or damage the cabling (See Specification 26.05.36 Section 3.4)

All the conduits mounted on the drywall in sales area, vestibule, amenity corridor and office area are concealed (See E-000 Note 21)

All the wall outlets shall be installed per plan, outlet boxes shall be installed at the level, and flash to the wall/tile



Outlet boxes are not aligned



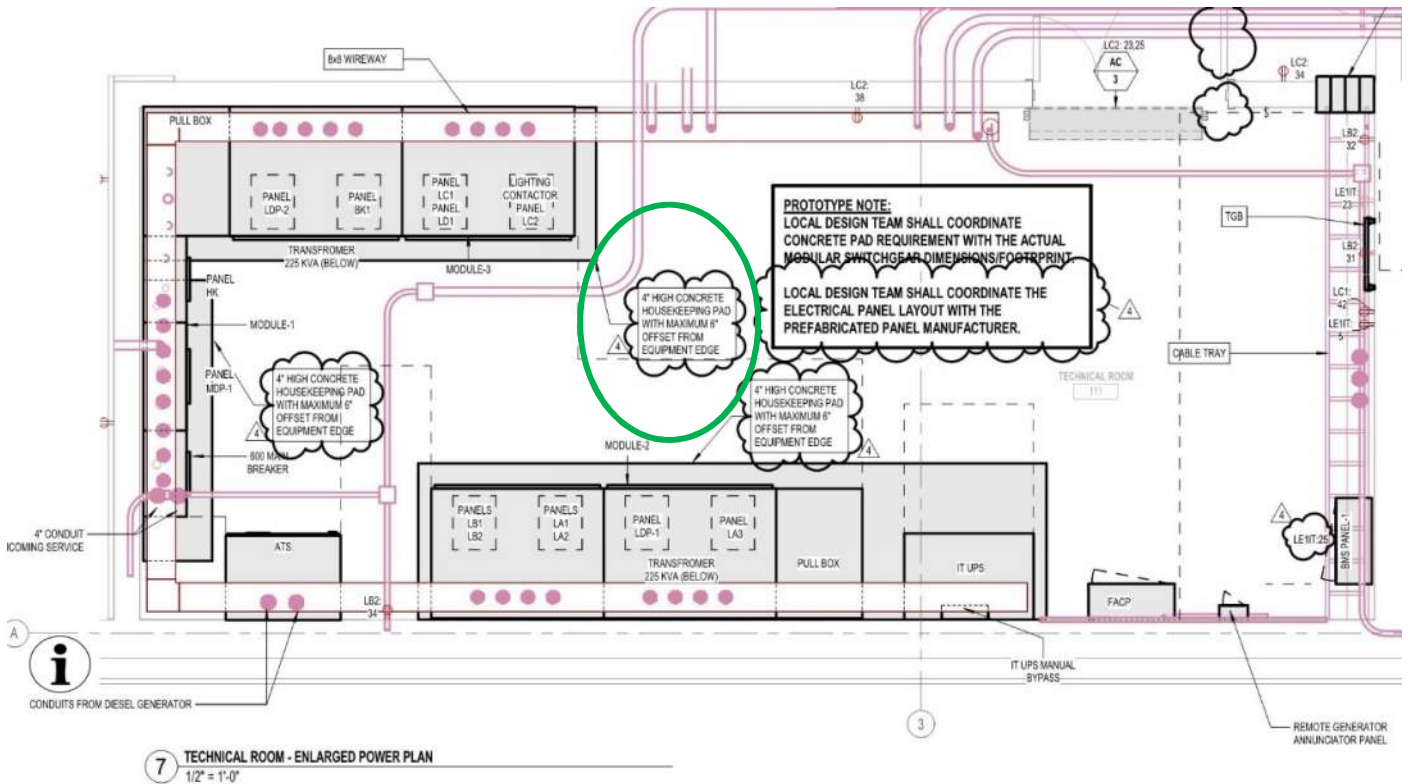
Outlet boxes are at the same level



Outlet covers are not flush with the surface

6.3.2 Switch Gears

All the switch gears shall be installed per approved plan (E-300) Orientation and the location of the equipment must be installed according to the approved drawings



Switch gears were not installed on the concrete pad

6.3.3 Panelboard

GC shall verify if the main breakers are installed per drawing – name of the panel, voltage, amperage, currents and lockouts (if applicable) must satisfies the design

PANEL BOARD: MDP-1													
LOCATION: TECHNICAL ROOM 111				VOLTS: 480/277 Wye				M.C. Rating: 65,000 AIC				NEUTRAL BUS: STANDARD, COPPER	
MOUNTING TYPE: PAD				PHASES: 3				MAINS TYPE: MAIN CB				GROUND BUS: STANDARD, COPPER	
ENCLOSURE: NEMA 1 MODULAR...				WIRES: 4				MAINS... 600 A					
FED FROM: UTILITY XFMR VIA ATS				MCB TYPE: COPPER				MCB RATING: 600 A					

CCT NO.	Load Name	TRIP	POLES	A		B		C		POLES	TRIP	Load Name	CCT NO.
1	PNL LDP-2 VIA TRANSFORMER T-1	350 A	3	65930 VA	56843 VA					3	350 A	PNL LDP-1 VIA TRANSFORMER T-2	2
3	--	--	--			64326 VA	57293 VA			--	--	--	4
5	--	--	--					65710 VA	51613 VA	--	--	--	6
7	RTU-1	60 A	3	13840 VA	49487 VA					3	225 A	PANEL HK	8
9	--	--	--			13840 VA	49487 VA			--	--	--	10
11	--	--	--					13840 VA	49487 VA	--	--	--	12
13	SURGE PROTECTION DEVICE	30 A	3	0 VA	0 VA					--	--	BREAKER SHUNT TRIP MODULE	14
15	--	--	--			0 VA	0 VA			--	--	SPACE	16
17	--	--	--					0 VA	0 VA	--	--	SPACE	18
TOTAL LOAD:				186100 VA		184947 VA		180650 VA					
TOTAL AMPS:				674 A		670 A		652 A					

LOAD CLASSIFICATIONS:	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	PANEL TOTALS
ELECTRICAL HEATING	32716 VA	100.00%	32716 VA	TOTAL CONNECTED LOAD: 549196 VA
HVAC	8830 VA	100.00%	8830 VA	TOTAL EST. DEMAND: 478685 VA
KITCHEN EQUIPMENT	0 VA	0.00%	0 VA	TOTAL CONN.: 661 A
LIGHTING	23759 VA	100.00%	23759 VA	TOTAL EST. DEMAND: 576 A
MOTOR/EQUIPT.	124576 VA	100.00%	124576 VA	
RECEPTACLE	60495 VA	58.27%	35247 VA	
REFRIGERATION	144960 VA	100.00%	144960 VA	
SPECIFIC RECEPTACLE	18020 VA	77.75%	14010 VA	
Lighting - Exterior	1500 VA	100.00%	1500 VA	
BAKERY EQUIPMENT	83760 VA	60.00%	50256 VA	
UPS RECEPTACLE	15500 VA	50.00%	7750 VA	

NOTES:
Provide A SHUNT-TRIP TYPE CIRCUIT BREAKER WITHIN MDP1 SERVING PANEL HK SO THAT IN THE EVENT OF A COMMERCIAL AC POWER FAILURE AND ACTIVATION OF THE FIRE PUMP VIA THE FIRE PUMP CONTROLLER, WHEN THE POWER IS FED FROM STAND-BY GENERATOR, THE 225A/3P CIRCUIT BREAKER SHALL OPEN VIA SHUNT TRIP.

GC shall verify if the shunt-trip breaker is installed and the shunt-trip is functioning correctly

ATS/Generator (specification 26.32.13; 26.36.00)

In order to provide a store with uninterrupted power supply a “Prototype 4” stores are equipped with the diesel-engine generator. After the completion of installation and during the acceptance of the store GC and CM shall verify if the tank is grounded and, FM shall confirm is it contains enough fuel for the generator to be in functionable condition for the duration of 72 hours

Automatic Transfer Switch must be installed per manufacture specification, and shall comply with approved drawings and **Specification 26.36.00**

6.3.4 Light fixtures

CM shall verify if the light fixtures were installed per light fixture schedule

F7	LED 36"	LED	80W 1100L	DRIVER	120V	PENDANT	OL LOPRETOP LED 11000-840 LB 001
F7 EV	LED 36" 16 MINUTE BATTERY PACK	LED	80W 1100L	DRIVER	120V	PENDANT	OL LOPRETOP LED 11000-840 LB 001 + EV LED DRIVER 8000NE 80L20M

F7 fixtures should be of 11000lums (light schedule). On site, the lights are only of 8000lums

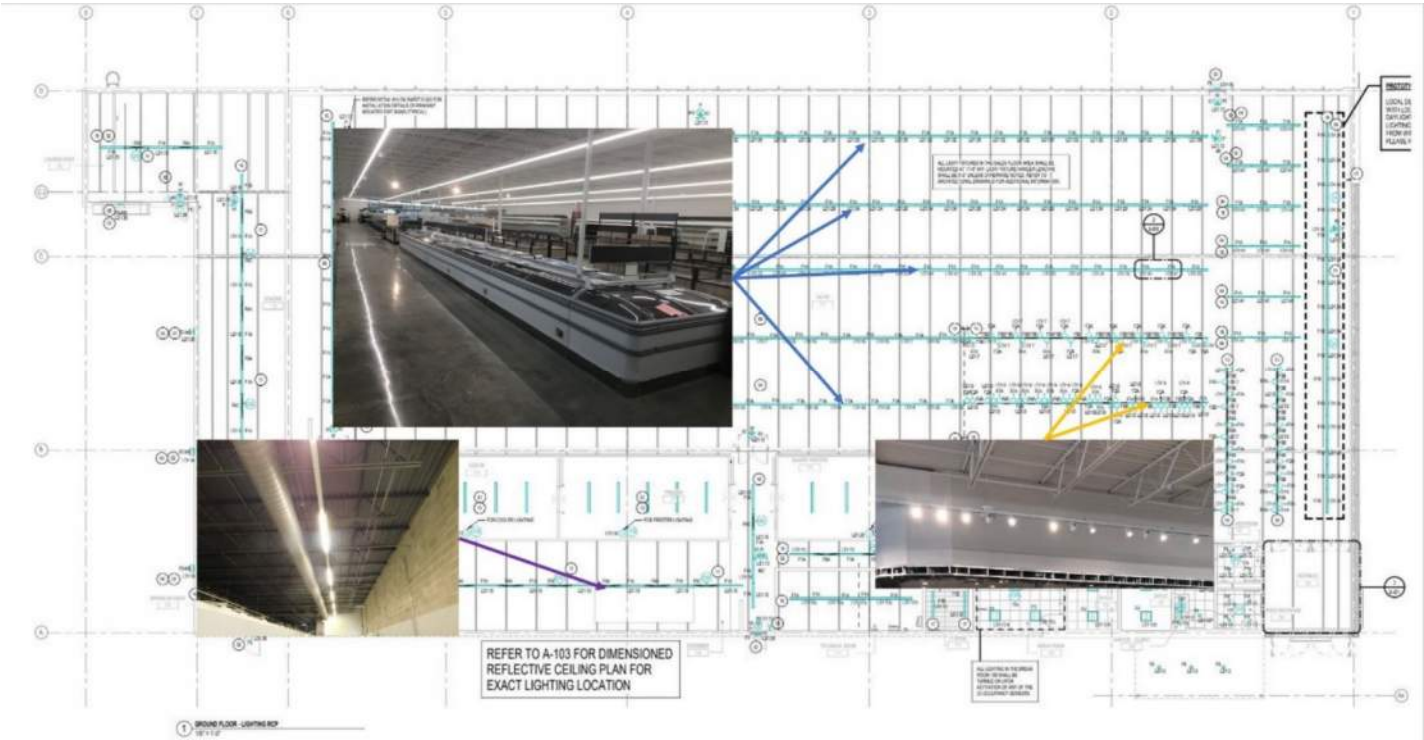
Light fixtures shall be installed as it specified in specification **26.51.00 section 3.1**

Square lights : at least 2 rods/wires

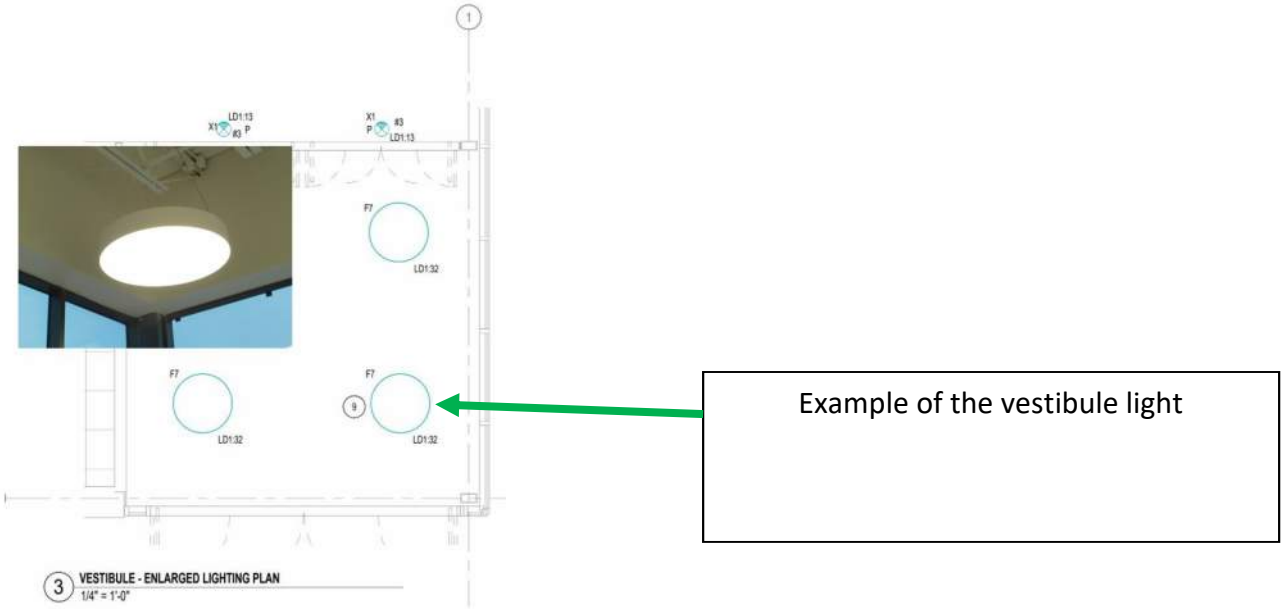
Single linear lights : twin-stem hangers

Continuous line : min one hanger per light

Location of the light fixtures



Vestibule lights

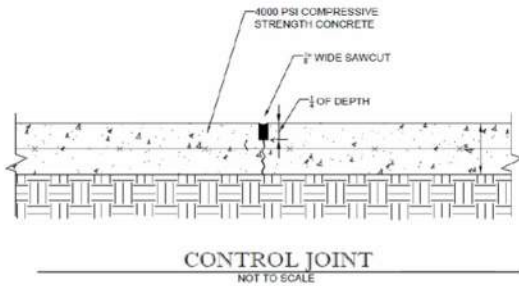


7. Finishes

7.1 Exterior

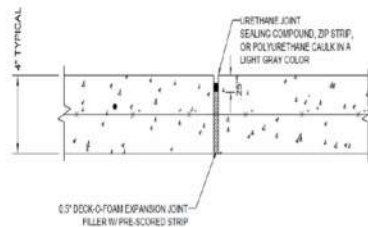
7.1.1 Concrete curb

For the correct functionality of the concrete its important to have the control joist installed per approved specification



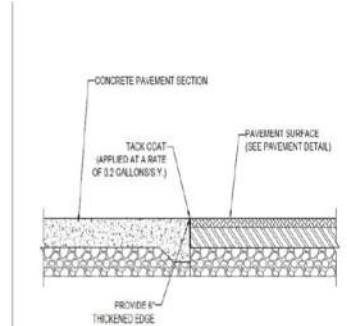
NOTES:

1. 1/2" WIDE, CONTROL JOINTS UTILIZING CONCRETE GROOVING TOOL AT THE FIRST PASS OR FLOATING, ONE QUARTER DEPTH OF SLAB.
2. PREFERRED CONCRETE CONTROL JOINT SPACING IS AT 6' INTERVALS FOR SIDEWALK AND 12' INTERVALS FOR LOADING DOCK AREA.



NOTES:

1. JOINT SPACING TO BE AS SHOWN ON PAGE 4



GC shall verify if the control joists are insta

EXPANSION JOINT

NOT TO SCALE

CONCRETE-TO-ASPHALT DETAIL

NOT TO SCALE



NOTES:

1. CONCRETE FOR CURBING SHALL HAVE A MINIMUM COMPRESSIVE STRENGTH OF 3,000 P.S.I. @ 28 DAYS.
2. CONSTRUCTION STAKING FOR CURB INSTALLATION SHALL BE REFERENCED (CUT OR FILL) TO THE TOP OF CURB.
3. AT CONTRACTOR'S OPTION, THE GUTTER THICKNESS MAY BE INCREASED AT THE EDGE OF PAVEMENT TO MAKE BOTTOM OF GUTTER CONTINUOUS WITH BOTTOM OF ASPHALT PAVEMENT.
4. CONTRACTION JOINTS SHALL BE PLACED @ 10'-0" O.C. TOOLED 1/4" (+/- 1/16") WIDE, 1" DEEP. EXPANSION JOINTS SHALL BE PLACED @ 40'-0" INTERVALS, MAXIMUM, AND ALL P.C.'S.
5. GUTTER PAN SLOPE TO BE ADJUSTED WITHIN ACCESSIBLE PARKING SPACES TO MATCH SLOPE BETWEEN SPOT ELEVATIONS.
6. GUTTER SLOPE TO MATCH ADJACENT PAVEMENT, TRAVERSE AND LONGITUDINAL.

CONCRETE CURB & GUTTER DETAIL

7.1.2 Asphalt

GC shall verify if the temperature of asphalt at the paving machine is between 270- and 320-degrees F

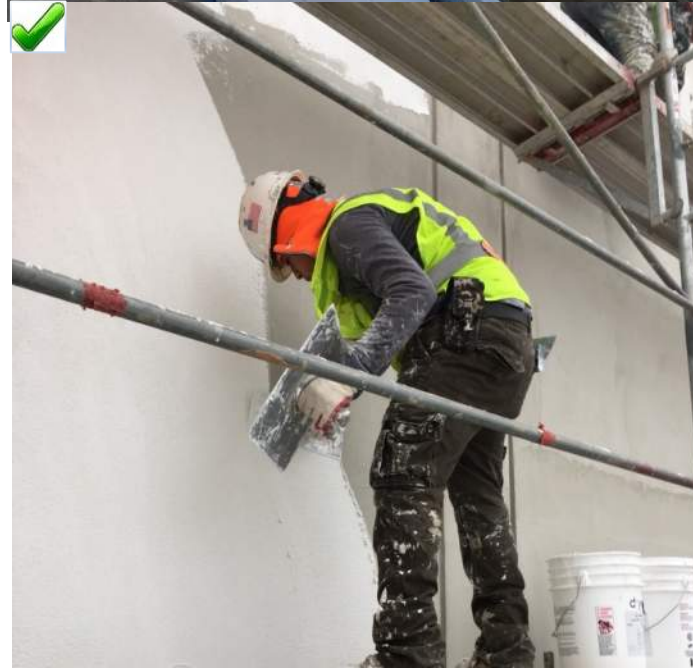
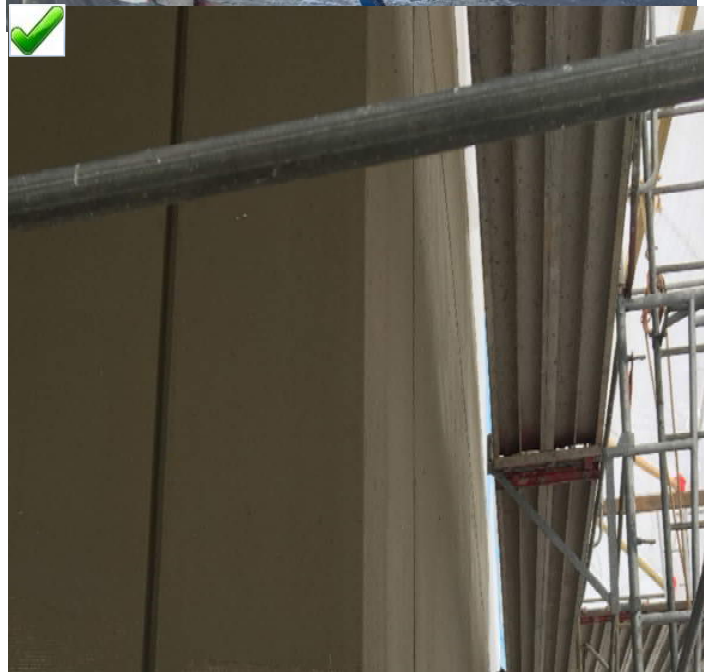


The finish rolling shall be completed while the material is workable to remove roller marks and leave a smooth pavement surface (compaction should be completed before the mix cools below 185 degrees F)



7.1.3 EIFS (Specification 07.24.19)

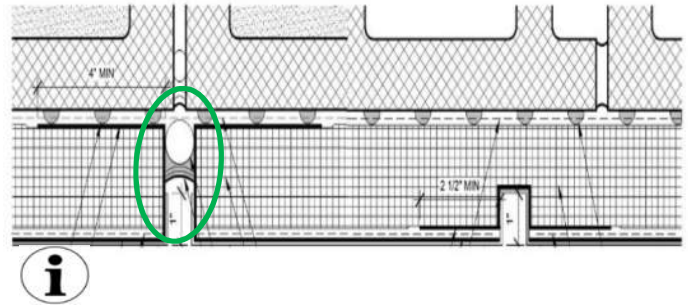
Exterior Insulation and Finish System (EIFS) with Moisture Drainage adhered over a fluid applied air and water-resistive barrier coating.



Sealant for decorative columns



Decorative column recesses sealant shall be installed at minimum of 1" from the surface



7.1.4 Brick veneer (Specification 04.00.00; 04.22.23)



Corner bricks shall be installed without any color indentations



GC shall verify the color of the brick veneer prior to installation
Surface shall be free from concrete stains

7.1.5 Split face CMU (Specification 04.00.00; 04.22.23)



Due to the nature of the split face CM Units, GC shall carefully select the pattern of the blocks

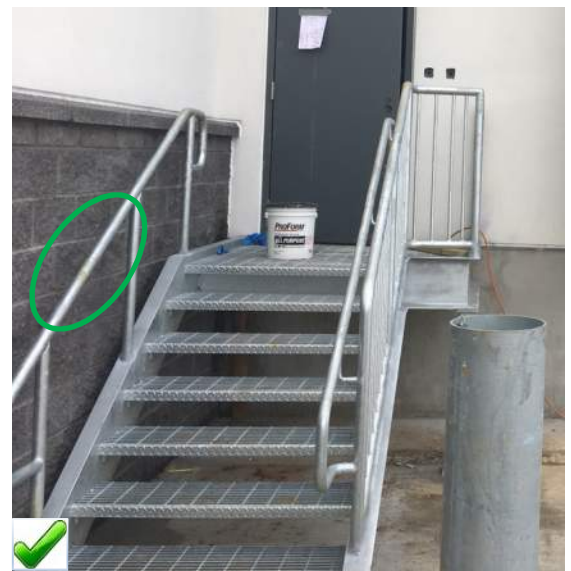


Clacking and brick mortar deviates from the specification.

7.1.6 Railing

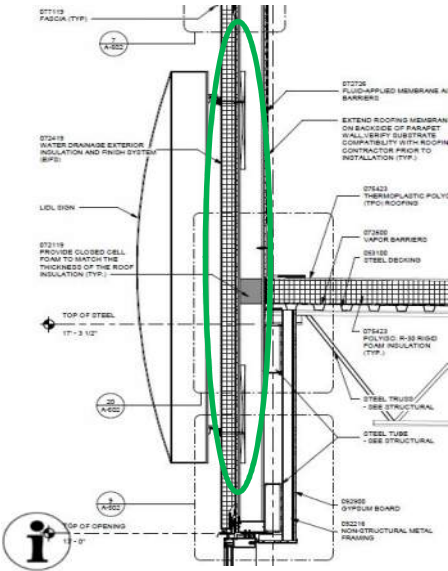


Railing shall be installed per approved drawings and shall comply with ADA requirements.

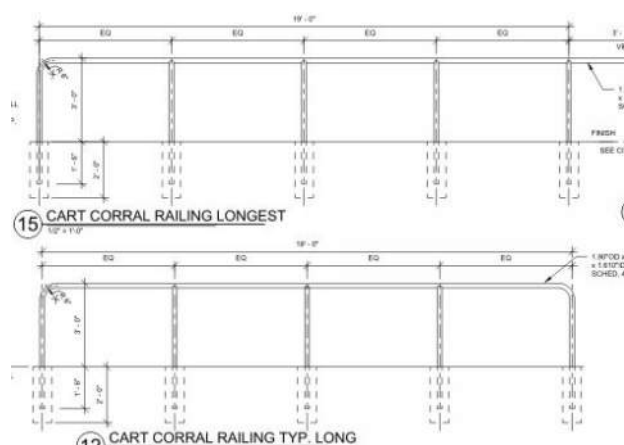


Correct installation of the railing

7.1.7 Store Signage



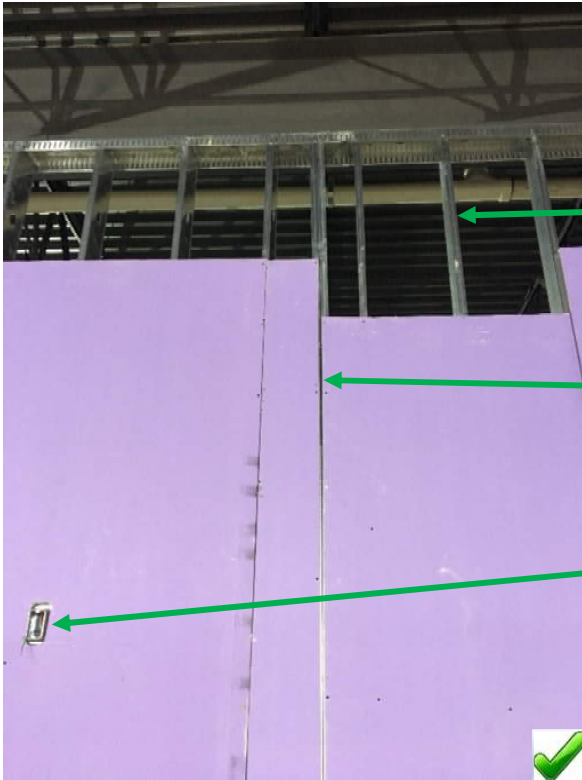
The cart coral railing shall be installed per approved drawings (A-001, A-400 and Specification Section 05.52.13)



7.2 Interior

7.2.1 Drywall /drywall finish

All the drywall installation shall be performed per approved drawings and must be strictly followed the floor elevations



All the drywall shall be installed on the metal studs

Control joints shall be installed with accommodation of the drywall type

Locations of the outlets shall be evaluated prior to installation



No drywall shall be installed on the surface of the store equipment nor on the areas without framing members

7.2.2 Paint

Each area has it designated paint or finish, therefore GC should follow the finish schedule. Moreover, some of the areas has two type of finish, for example restrooms

ROOM FINISH SCHEDULE																	
ROOM NO.	ROOM NAME	Area	FLOOR		BASE		WALLS								CEILINGS		
			MAT'L	FIN	MAT'L	FIN	BACK	RIGHT	FRONT	LEFT	MAT'L	FIN	MAT'L	FIN	MAT'L	FIN	HEIGHT
100	VESTIBULE	325 SF	CONC	F1	-	-	GYP	P1	GYP	P1	GYP	P1	GYP	P1	ES-1	-	VARIES
101	SALES	15453 SF	CONC	F1	B1	FF	GYP	P1	GYP	P1	GYP	P1	GYP	P1	ES-1	-	VARIES
102	BAKERY	613 SF	CONC	F5	B2	SS	GYP	P4	GYP	P4	GYP	FRP,SS	GYP,IMP	P4,FF	ACT-2	FF	10' - 0"
103	UNISEX RESTROOM	69 SF	PT	FF	-	-	GYP	PT,P1	GYP	PT,P1	GYP	PT,P1	GYP	PT,P1	ACT-1	FF	8' - 10"
104	UNISEX RESTROOM	69 SF	PT	FF	-	-	GYP	PT,P1	GYP	PT,P1	GYP	PT,P1	GYP	PT,P1	ACT-1	FF	8' - 10"

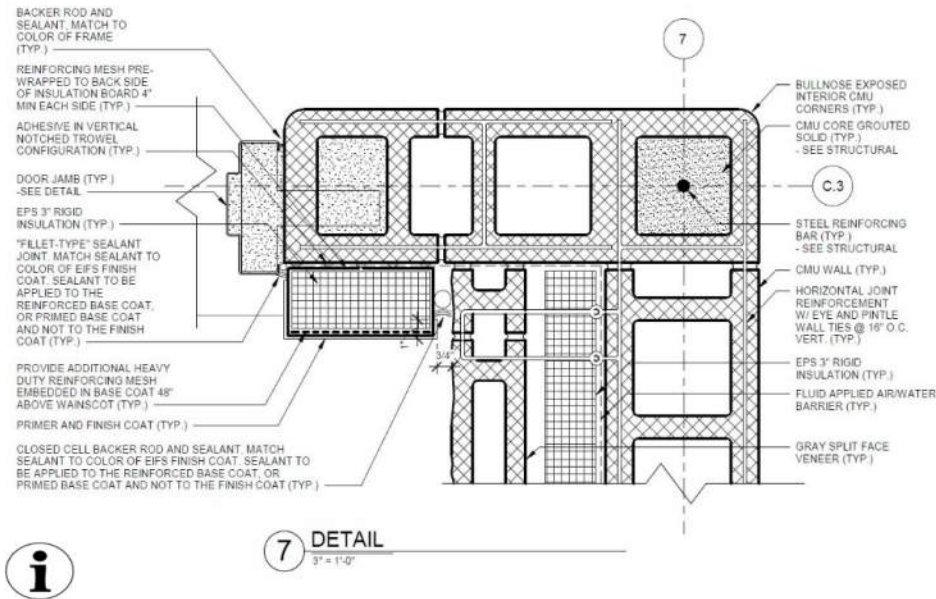
After the completion of the work, CM shall verify if the shade of the color matches the design and, in some cases, GC shall leave the paint cans after completion of the work.

PAINT FINISHES				
PAINT FINISHES	FF	FACTORY FINISH		
PAINT FINISHES	P0	UNPAINTED		
PAINT FINISHES	P1	PAINT	SHERWIN WILLIAMS	EGGSHELL
PAINT FINISHES	P2	PAINT	SHERWIN WILLIAMS	EGGSHELL
PAINT FINISHES	P3	PAINT	SHERWIN WILLIAMS	SEMIGLOSS
PAINT FINISHES	P4	PAINT	SHERWIN WILLIAMS	EPOXY PAINT EGGSHELL
PAINT FINISHES	P5	PAINT	SHERWIN WILLIAMS	EGGSHELL
PAINT FINISHES	P6	PAINT	SHERWIN WILLIAMS	FLAT

7.2.3 Door opening

All the door opening shall be installed with a bullnose edge

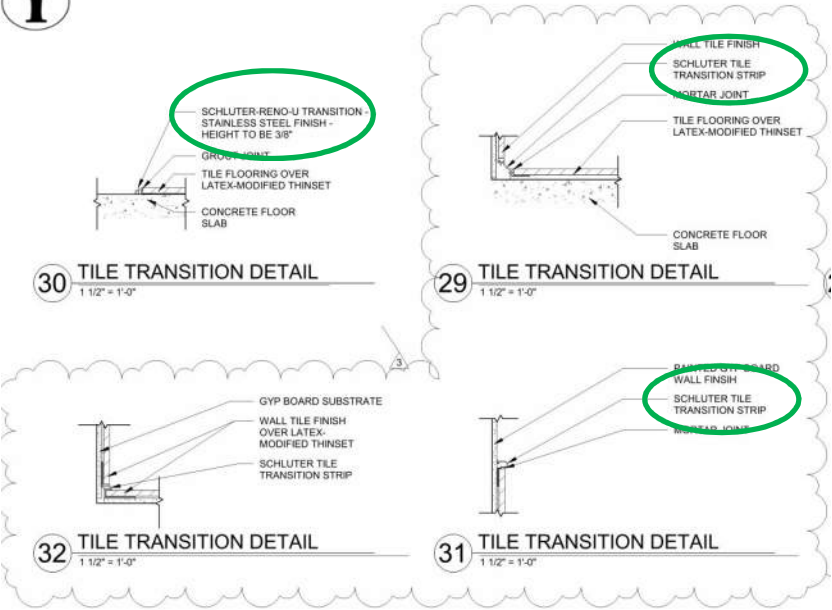




7.2.4 Tile (Specification 09.30.13)

Tile shall be installed only in the areas where is specified on the drawings – currently prototype 4 store has tile in the restrooms and on the scrubbing station walls. Tiles shall be installed in the staggered pattern.





All the edges shall be finished with a Schluter tile transitions



8. Enforcing quality assurance

8.1 Checklists

Checklists app is designed to help Construction Managers to control the quality of the construction.

Quality Department has issued 7 seven checklists: Foundation, Slab, Structural, Mechanical, Electrical, Underground MEP and Exterior Finish

Each of the checklist shall be completed by the Construction Managers at the appropriate period time frame during the construction. And shall be verified by the Quality Manager

8.2 Time line



8.3 Examples of the check lists

If Construction Manager experiences a problem with accessing Checklists en App a .pdf format can be used. However, Quality Departments strongly recommends to use checklists through the app.



Checklist for stores construction management

January 2019

Foundation trade

Pre-Construction Meeting

#	Topic	Reference	Yes / No
1	Have all submittals been validated ?		
2	Have soil pressure capacity inspection been completed and net bearing capacity of 3000PSF requirement been met ?	S405 - Foundation Schedule, note 3	
3	Do the top of foundation require reinforcement ?	S405 - Foundation Schedule, note 3	
4	Have there been discussion about the QA of the GC ? (concrete tracking, dewatering management, etc.)		
5	Has there been a discussion about the construction checklist ?		
6	Is the generator foundation size compatible with the generator shop drawings ?	A-002 - detail 5	
7	Has the transformer foundation size and location been validated with the utility company ?	A-005	

Construction Checklist

#	Topic	Reference	Yes / No
1	Have the concrete samples been completed and monitored? (1/2 samples on site + 1/2 samples on laboratory, 7 days and 28 days compression test)	0033000 3, L2.D.	
2	Are the foundation layouts done by a geometer ?		





Pdf format – can be print out prior to the site visit.

Structure

All

Open

Comment

Defective

Header data

Construction Manager

Stephan Darmio

General Contractor

Irwin Kamp; Lighthouse

Comment

Comment

Pre-Construction Meeting

Construction

Masonry

Structural Steel

Roof Decking

Was the decking installed perpendicular to the joists?

Comment

Incomplete at time of inspection, will follow up at next site visit.

Defective

X

Checklisten app can be accessed by Cm at any time



9. Prototype 4 modification – T21, T17, T14